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# LECTURE 6: Lennard-Jones Fluid Pair Potential
# Script: LJ
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# Description:
#   This notebook lets you interactively adjust epsilon and
↪   sigma of the LJ
#   potential and see how it changes the shape of the potential.
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import sys

# Check if running in Colab and install ipympl for interactivity
↪ if needed
# (ipywidgets is usually pre-installed, but this ensures
↪ matplotlib works nicely)
try:
    import google.colab
    IN_COLAB = True
    print("Running in Google Colab. Installing ipympl...")
    %pip install ipympl
    from google.colab import output
    output.enable_custom_widget_manager()
except ImportError:
    IN_COLAB = False
    print("Running locally.")

import numpy as np
import matplotlib.pyplot as plt
from ipywidgets import interact, FloatSlider
import ipywidgets as widgets

```

Running locally.

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def plot_lennard_jones(epsilon_k, sigma):
    """
    Plots the Lennard-Jones potential.
    epsilon_k: Depth of the potential well in Kelvin (epsilon /
↪ kB)
    sigma: Distance where potential is zero (nm)
    """

    # 1. Define the range of r (distance)
    # We choose a range that covers most small molecules (0.25nm
↪ to 1.0nm)
    r = np.linspace(0.24, 1.0, 500)

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# 2. Calculate Potential Energy (in Kelvin)
#  $V(r) = 4 * \epsilon * [ (\sigma/r)^{12} - (\sigma/r)^6 ]$ 
# We plot  $V/k_B$ , so we use epsilon_k directly
V_over_kB = 4 * epsilon_k * ((sigma / r)**12 - (sigma /
↪ r)**6)

# 3. Create Plot
plt.figure(figsize=(8, 6))
plt.plot(r, V_over_kB, linewidth=3, color='blue', label='LJ
↪ Potential')

# 4. Mark the Minimum (Equilibrium separation)
# The minimum occurs at  $r = 2^{(1/6)} * \sigma$ 
r_min = (2**(1/6)) * sigma
V_min = -epsilon_k

plt.plot(r_min, V_min, 'ro', markersize=10,
        label=f'Well Depth ( $\epsilon/k_B$ ) =  $-\epsilon_k$ 
↪ K')

# 5. Formatting
plt.axhline(0, color='black', linestyle='--', linewidth=1)
plt.ylim(-400, 200) # Fixed Y-limit to allow visual
↪ comparison of well depths
plt.xlim(0.24, 1.0)
plt.xlabel('Intermolecular Distance  $r$  (nm)', fontsize=12)
plt.ylabel('Potential Energy /  $k_B$  (K)', fontsize=12)
plt.title(f'Lennard-Jones Potential:  $\epsilon/k_B =$ 
↪ {epsilon_k:.0f} K,  $\sigma =$  {sigma:.3f} nm', fontsize=14)
plt.grid(True, which='both', linestyle='--', alpha=0.5)
plt.legend(fontsize=11)

# N. B. T_boiling is roughly  $0.7 * \epsilon/k_B$  for noble
↪ gases

plt.show()

# Create the sliders
# Ranges cover Hydrogen (small) to Xenon (large)
interact(plot_lennard_jones,
        epsilon_k=FloatSlider(value=120, min=30, max=300,
↪ step=10,
                                description='epsilon/kB (K):'),
        sigma=FloatSlider(value=0.34, min=0.25, max=0.45,
↪ step=0.01,
                                description='sigma (nm):'));

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interactive(children=(FloatSlider(value=120.0, description='epsilon/kB (K) :
```